Stars and bars, inclusion-exclusion and bijections

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L3-CO1 Level 3 Senior · Combinatorics

Count distributions with stars and bars, repair overcounts with inclusion-exclusion, and turn awkward counts into easy ones with bijections.
By the end you can count distributions of identical objects, with lower and upper limits, using stars and bars; count objects avoiding several bad events with inclusion-exclusion; replace an awkward count by an easy one using a bijection, such as squeezing gaps.
Prerequisites: Counting without missing anything (L2-CO1), Factors, primes and divisibility (L2-NT1). **Contests:** SMC, Senior Kangaroo, Maclaurin, AMC 12, AIME. **Time:** about 90 minutes.

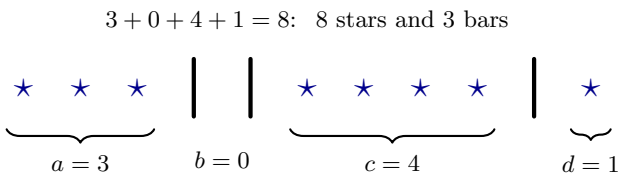
1 Lesson

Consider the number of ways to write 10 as an ordered sum $a+b+c$ of non-negative integers, where $3+0+7$ and $0+3+7$ count as different. Listing $(10+0+0,\ 9+1+0,\ 9+0+1,\ \ldots)$ is slow and error-prone, and a contest would ask about 40 rather than 10. The answer is $\binom{12}{2} = 66$, by the method of the first section.

This unit covers three methods. *Stars and bars* counts the ways to share out identical things. *Inclusion-exclusion* repairs overcounting when conditions overlap. *Bijections* let you count an awkward set by pairing off its elements exactly with the elements of a set that is easy to count. Between them they cover a large share of the counting questions on the SMC, the Senior Kangaroo and the early AIME.

1.1 Stars and bars

Write each solution of $a+b+c=10$ as a row of 10 stars cut into three groups by 2 bars. For instance $3+0+7$ becomes three stars, a bar, nothing, a bar, then seven stars. Every row of 10 stars and 2 bars, in any order, gives exactly one solution (read off the sizes of the three groups), and every solution gives exactly one row. So we only need to count rows. There are 12 positions and we choose which 2 of them hold bars, giving $\binom{12}{2} = 66$. The picture shows the row for $3+0+4+1$, one of the solutions of $a+b+c+d=8$.



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Stars and bars.
The number of ordered k-tuples of non-negative integers $(x_1, \ldots, x_k)$ with $x_1 + x_2 + \cdots + x_k = n$ is

$$\binom{n+k-1}{k-1}.$$

If every x_i must be at least 1, the number is $\binom{n-1}{k-1}$.

The proof of the first formula is the picture. A solution is a row of n stars and $k-1$ bars, and such a row is fixed by choosing which $k-1$ of the $n+k-1$ positions are bars. For the second, either hand one star to each variable first, leaving $n-k$ to share freely in $\binom{n-1}{k-1}$ ways, or line up the n stars and choose $k-1$ of the $n-1$ gaps between them as cutting points. Both arguments are worth knowing, because they generalise in different directions.

Two habits make the formula more useful. First, lower bounds are removed by substitution. If $x_1 \geq 3$, write $x_1 = y_1 + 3$ with $y_1 \geq 0$, and the total drops by 3. Second, an inequality becomes an equation with a slack variable. Solutions of $x_1 + \cdots + x_k \leq n$ match up with solutions of $x_1 + \cdots + x_k + s = n$ where $s \geq 0$, so there are $\binom{n+k}{k}$ of them.

The model is n *identical* objects going into k *labelled* boxes. If the objects are all different, each one simply picks a box and there are k^n ways. If the boxes are unlabelled, you are counting partitions, there is no neat formula, and you must organise a careful list.

Example 1.
A bakery sells four kinds of bun, namely plain, cinnamon, raisin and chocolate. Mia buys 12 buns, with at least one of each kind and at least three cinnamon buns. How many different selections are possible?

Solution.
Buns of the same kind are identical, so a selection is a list of four numbers (p, c, r, h) adding up to 12. Splitting into cases on the number of cinnamon buns is slow and a case is easily dropped, so we deal with the lower bounds first.
Put one plain, three cinnamon, one raisin and one chocolate bun in the bag before choosing anything. That uses 6 buns, and the other 6 can now be shared among the four kinds with no restriction at all. So we want non-negative solutions of $p' + c' + r' + h' = 6$, and stars and bars gives $\binom{9}{3} = 84$ selections.

1.2 Inclusion-exclusion

Stars and bars does not handle upper bounds directly, so for these we use inclusion-exclusion.

Inclusion-exclusion.
For finite sets $A_1, \ldots, A_m$,

$$|A_1 \cup \cdots \cup A_m| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<l} |A_i \cap A_j \cap A_l| - \cdots + (-1)^{m+1} |A_1 \cap \cdots \cap A_m|.$$

So the number of elements of a universe U lying in *none* of the A_i is $|U|$ minus the first sum, plus the second, minus the third, and so on.

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Proof. Take an element lying in exactly $r \geq 1$ of the sets. This element is counted $\binom{r}{1}$ times in the first sum, $\binom{r}{2}$ times in the second, and so on, so its total contribution is

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots \pm \binom{r}{r} = 1 - (1-1)^r = 1$$

by the binomial theorem. An element in none of the sets contributes nothing. So the right-hand side counts each element of the union exactly once.

In practice the A_i are "bad events" and you want the objects with no bad event at all. The method is efficient when the intersections are easy to count, and especially when all intersections of the same number of sets have equal size, so that each sum is a binomial coefficient times one representative term. Check for this symmetry before you start.

Example 2.
A quiz has four rounds, and each round is marked out of 7 in whole numbers. In how many ways can a team score 20 in total? (A way means a list of four round scores.)

Solution.
Without the upper limit this is $a+b+c+d = 20$ in non-negative integers, which has $\binom{23}{3} = 1771$ solutions. This overcounts, since this count includes solutions in which some round scores 8 or more. Let A_i be the set of solutions in which round i scores at least 8.
For $|A_1|$, write $a = a' + 8$, so that $a' + b + c + d = 12$, giving $\binom{15}{3} = 455$. For $|A_1 \cap A_2|$, take 8 off two scores, so that the four numbers add to 4, giving $\binom{7}{3} = 35$. Three rounds of 8 or more would need at least 24 marks, so the triple intersections are empty. By symmetry every single set has size 455 and every pair has size 35, so the number of good solutions is

$$1771 - 4 \cdot 455 + 6 \cdot 35 = 1771 - 1820 + 210 = 161.$$

Note that $4 \cdot 455$ is bigger than 1771, so the running total is negative after the first correction and becomes positive again only with the pair term. This is not an error, and the negative total shows why you must never stop after the first term.
There is a quick check. Replacing each score x by $7 - x$ is a bijection from totals of 20 to totals of 8, and with a total of 8 the only bad solutions are those with a single round on 8 and the rest on 0. So the count is $\binom{11}{3} - 4 = 165 - 4 = 161$, which agrees.

1.3 Bijections

A bijection between two sets is a pairing in which every element of one is paired with exactly one element of the other. If you can build one from an awkward set to a set you know how to count, you are done. Stars and bars is already a bijection (solutions matched with rows of symbols), and so was the check at the end of Example 2. The next result is another standard bijection.

Spread-out choices.
The number of ways to choose k numbers from $\{1, 2, \ldots, n\}$ so that any two chosen numbers differ by at least d is

$$\binom{n - (k-1)(d-1)}{k}.$$

With $d = 2$ (no two consecutive) this is $\binom{n-k+1}{k}$.

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Proof. Write the chosen numbers as $a_1 < a_2 < \cdots < a_k$ with $a_{i+1} - a_i \geq d$, and set $b_i = a_i - (i-1)(d-1)$. Then $b_{i+1} - b_i = a_{i+1} - a_i - (d-1) \geq 1$, so the b_i are increasing, and $b_1 = a_1 \geq 1$ while $b_k \leq n - (k-1)(d-1)$. So $\{b_1, \ldots, b_k\}$ is an ordinary k-element subset of $\{1, \ldots, n-(k-1)(d-1)\}$. Conversely, starting from any such subset and adding $(i-1)(d-1)$ back to its i-th smallest element gives a valid choice of a's. The two maps undo each other, so the two sets have the same size.

A bijection proof must show that the map lands in the right set *and* that the map can be reversed, as the last three sentences of the proof do. Students often check only the first, and then the map may miss or double-count some objects.

Example 3.
In how many ways can you choose 4 numbers from $\{1, 2, \ldots, 20\}$ so that any two of them differ by at least 3?

Solution.
One could count the complement, the sets with some pair differing by 1 or 2, but those bad events overlap in many ways (one set can contain several close pairs), and the inclusion-exclusion is long. When the condition is about spacing, squeeze the gaps instead (the spread-out choices bijection).
Write the choice as $a_1 < a_2 < a_3 < a_4$ and slide the numbers together by setting $b_1 = a_1$, $b_2 = a_2 - 2$, $b_3 = a_3 - 4$, $b_4 = a_4 - 6$. Each consecutive difference drops by 2, so every difference is still at least 1, and $b_4 \leq 14$. For instance $\{2, 5, 11, 20\}$ becomes $\{2, 3, 7, 14\}$. Every 4-element subset of $\{1, \ldots, 14\}$ arises exactly once, because we can always spread the numbers back out, so the answer is $\binom{14}{4} = 1001$.
The same count also follows from stars and bars. Let $g_0 = a_1 - 1$, let g_1, g_2, g_3 be the numbers of unchosen integers between consecutive choices, and let $g_4 = 20 - a_4$. Then $g_0 + g_1 + g_2 + g_3 + g_4 = 16$ with $g_1, g_2, g_3 \geq 2$. Taking 2 off each middle gap leaves 10 shared among 5 gaps, and $\binom{14}{4} = 1001$ again.

Example 4.
How many arrangements of the seven letters A, A, B, B, C, C, D have no two identical letters next to each other?

Solution.
A direct approach places letters one at a time, but the number of options for each place depends on what came before, so the cases multiply quickly. The condition says that three events must not happen, so we use inclusion-exclusion.
There are $\frac{7!}{2!\,2!\,2!} = 630$ arrangements in total. Let A be the set where the two A's are adjacent, and define B and C in the same way. To count A, glue the two A's into one tile. Arrangements with the A's adjacent match up with arrangements of the six objects AA, B, B, C, C, D, of which there are $\frac{6!}{2!\,2!} = 180$. This matching is a bijection because the A's are identical, so the tile has only one internal order. (With distinguishable letters you would multiply by 2.) Gluing two pairs leaves five objects with one repeated letter, giving $\frac{5!}{2!} = 60$, and gluing all three leaves $4! = 24$. By symmetry,

$$630 - 3 \cdot 180 + 3 \cdot 60 - 24 = 630 - 540 + 180 - 24 = 246.$$

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Common traps.

- Stars and bars needs identical objects and labelled boxes. Different objects give k^n, while unlabelled boxes need a careful list.
- Mixing up the two formulas, where $\binom{n+k-1}{k-1}$ allows zeros and $\binom{n-1}{k-1}$ does not. Test on a tiny case, such as $n = 2$, $k = 2$, if in doubt.
- Stopping inclusion-exclusion after one term, getting a sign wrong, or forgetting that some intersections are empty.
- Counting multiples of 4 and 6 by multiples of 24, forgetting that the intersection is multiples of the lowest common multiple, 12.
- Forgetting in digit problems that each digit is at most 9 and the leading digit is at least 1.
- Calling a map a bijection without checking that the map can be reversed.

1.4 How to spot this in a contest

The signals for each method are as follows.

1. The number of ways to share, split or distribute something identical, or the number of integer solutions of a sum, so use stars and bars, with lower bounds handled by giving out the minimum first. Phrases like "at most", "no more than" or a digit sum mean upper bounds, so add inclusion-exclusion.
2. "None of", "no two of the same", "every one of them at least once", or several conditions that must all fail, so count the complement with inclusion-exclusion, and check that the intersections are symmetric and easy to count.
3. A condition about gaps, order or spacing ("no two adjacent", "differ by at least", "increasing"), so look for a bijection that removes the condition, by squeezing the gaps, recording the differences, or turning an increasing sequence into a subset.

In multiple-choice papers, a useful test is to try the smallest non-trivial case by hand and see which formula fits.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ A shop sells 6 kinds of sticker, with plenty of each kind. Zara buys a pack of 10 stickers containing at least one sticker of each kind. How many different packs are possible? (Two packs are the same if they contain the same number of each kind.)

Problem 2 ⋆⋆ How many integers from 1 to 2000 inclusive are divisible by at least one of 4, 6 and 10?

Problem 3 ⋆⋆ How many three-digit numbers have digits adding up to 15?

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Problem 4 ⋆⋆⋆ Twenty people sit around a circular table. In how many ways can four of them be chosen so that no two of the chosen people are sitting next to each other?

Problem 5 ⋆⋆⋆ Eight different prizes are given to four students, Ana, Ben, Cal and Dev, so that each student receives at least one prize. In how many ways can this be done?

Problem 6 ⋆⋆⋆⋆ A bag contains 8 red and 8 blue balls. They are taken out one at a time and their colours are written down in order. In how many of the possible colour sequences is the number of red balls taken out always at least the number of blue balls taken out, at every stage?

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT Senior Kangaroo 2023 Question 12** ⋆
   Each such number is an ordered split of 6 into positive parts
2. **AMC 12 2001 Problem 12** ⋆
   Inclusion-exclusion with lowest common multiples
3. **UKMT SMC 2021 Question 15** ⋆⋆
   Stars and bars, then remove the solutions where a die exceeds 6
4. **AMC 10A 2018 Problem 11** ⋆⋆
   Shift each die down so that zeros are allowed, then check whether any upper limit can bite
5. **AIME II 2017 Problem 1** ⋆⋆
   Complement, then two-set inclusion-exclusion
6. **AIME 1998 Problem 7** ⋆⋆
   Substitute to remove the oddness condition
7. **UKMT BMO1 2009/10 Problem 3** ⋆⋆⋆
   A non-increasing sequence is fixed by how many times each mark occurs
8. **AIME II 2009 Problem 6** ⋆⋆⋆
   Count the complement by squeezing the gaps
9. **UKMT BMO1 2010/11 Problem 4** ⋆⋆⋆
   Pair up arrangements by changing one well-chosen counter
10. **AIME I 2021 Problem 4** ⋆⋆⋆
    Rewrite the piles in terms of differences to remove the inequalities
11. **AIME II 2001 Problem 9** ⋆⋆⋆
    Inclusion-exclusion over the four 2 by 2 blocks
12. **UKMT BMO1 2014/15 Problem 3** ⋆⋆⋆⋆
    Split by how many rooms are on each side, then count non-adjacent choices

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4 Hints

1. Put one sticker of each kind in the pack first. What is left to decide?
2. Use inclusion-exclusion, and be careful about which numbers are divisible by both 4 and 6.
3. Write the number as $\overline{abc}$ and shift a so that all three variables start at 0. Which digits could break the upper limit of 9?
4. Number the seats 1 to 20. Count as if the seats were in a row, then deal with the one pair of seats that are neighbours only because the table is round.
5. The prizes are different, so stars and bars does not apply. Count all ways first, then subtract the ways in which some student gets nothing.
6. Count the bad sequences. At the first moment the blues get ahead, swap the colours of everything taken out so far. What kind of sequence do you get?

5 Solutions

Problem 1

A shop sells 6 kinds of sticker, with plenty of each kind. Zara buys a pack of 10 stickers containing at least one sticker of each kind. How many different packs are possible? (Two packs are the same if they contain the same number of each kind.)

A pack is determined by the numbers $(x_1, \ldots, x_6)$ of stickers of each kind, with $x_1 + \cdots + x_6 = 10$ and every $x_i \geq 1$. Put one sticker of each kind in first, and the remaining 4 stickers can be shared among the 6 kinds with no restriction. By stars and bars this gives $\binom{4+5}{5} = \binom{9}{5} = 126$ packs. (Equivalently, $\binom{10-1}{6-1} = 126$ by the formula for positive solutions.)

Answer: 126

Problem 2

How many integers from 1 to 2000 inclusive are divisible by at least one of 4, 6 and 10?

Let A, B and C be the sets of multiples of 4, 6 and 10 up to 2000. Then $|A| = 500$, $|B| = \lfloor 2000/6 \rfloor = 333$ and $|C| = 200$. A number divisible by both 4 and 6 is a multiple of their lowest common multiple, which is 12, not 24. So $|A \cap B| = \lfloor 2000/12 \rfloor = 166$, and similarly $|A \cap C| = \lfloor 2000/20 \rfloor = 100$ and $|B \cap C| = \lfloor 2000/30 \rfloor = 66$. Finally $A \cap B \cap C$ is the set of multiples of 60, and $|A \cap B \cap C| = \lfloor 2000/60 \rfloor = 33$.

By inclusion-exclusion, the count is $500 + 333 + 200 - 166 - 100 - 66 + 33 = 734$.

Answer: 734

Problem 3

How many three-digit numbers have digits adding up to 15?

Write the number as $\overline{abc}$ with $1 \leq a \leq 9$ and $0 \leq b, c \leq 9$. Put $a = a' + 1$, so we need $a' + b + c = 14$ with $0 \leq a' \leq 8$ and $0 \leq b, c \leq 9$. Without the upper limits there are $\binom{16}{2} = 120$ solutions.

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Now remove the bad ones. If $a' \geq 9$, write $a' = a'' + 9$ to get $a'' + b + c = 5$, which has $\binom{7}{2} = 21$ solutions. If $b \geq 10$, write $b = b' + 10$ to get $a' + b' + c = 4$, which has $\binom{6}{2} = 15$ solutions, and the same holds for $c \geq 10$. Any two of these at once would need a total of at least 19, which is more than 14, so no solution is bad in two ways. Hence the count is $120 - 21 - 15 - 15 = 69$.

Answer: 69

Problem 4

Twenty people sit around a circular table. In how many ways can four of them be chosen so that no two of the chosen people are sitting next to each other?

Number the seats 1 to 20 around the table. If the seats were in a row, the number of ways to choose 4 with no two consecutive would be $\binom{20-4+1}{4} = \binom{17}{4} = 2380$, by the spread-out choices formula. Around the table, seats 1 and 20 are also neighbours, so we must remove the row choices that contain both 1 and 20.

If both 1 and 20 are chosen, then seats 2 and 19 cannot be, and the other two chosen seats form a non-consecutive pair from $\{3, 4, \ldots, 18\}$, which has 16 seats. There are $\binom{16-2+1}{2} = \binom{15}{2} = 105$ such pairs. So the answer is $2380 - 105 = 2275$.

Answer: 2275

Problem 5

Eight different prizes are given to four students, Ana, Ben, Cal and Dev, so that each student receives at least one prize. In how many ways can this be done?

Since the prizes are different, each prize independently chooses a student, so there are 4^8 ways with no restriction. Let A_i be the set of ways in which student i gets nothing. Then $|A_i| = 3^8$, since each prize chooses among the other three students, $|A_i \cap A_j| = 2^8$, $|A_i \cap A_j \cap A_k| = 1$, and all four cannot be empty-handed. By inclusion-exclusion the number of ways in which everyone gets a prize is

$$4^8 - \binom{4}{1}3^8 + \binom{4}{2}2^8 - \binom{4}{3}1^8 = 65536 - 26244 + 1536 - 4 = 40824.$$

Answer: 40824

Problem 6

A bag contains 8 red and 8 blue balls. They are taken out one at a time and their colours are written down in order. In how many of the possible colour sequences is the number of red balls taken out always at least the number of blue balls taken out, at every stage?

A colour sequence is a string of 8 R's and 8 B's, so there are $\binom{16}{8} = 12870$ sequences in all. Call a sequence bad if at some stage there are more B's than R's so far. In a bad sequence, look at the first such stage. At that point there are j R's and $j + 1$ B's for some j. Swap the colours of all the balls up to and including that point. The new sequence has $j + 1$ R's and j B's at that point, and is unchanged after that point, so the new sequence contains 9 R's and 7 B's in total.

This map is reversible. In any sequence of 9 R's and 7 B's, the excess of R's over B's goes from 0 to 2 one step at a time, so there is a first stage at which there is exactly one more R than B. Swapping the colours up to that stage gives a sequence of 8 R's and 8 B's whose first stage with more B's

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than R's is that same stage, so we recover a bad sequence, and the two maps undo each other. Hence the bad sequences are in bijection with the sequences of 9 R's and 7 B's, of which there are $\binom{16}{7} = 11440$.

The number of good sequences is $12870 - 11440 = 1430$.

Answer: 1430

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